HW5

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statement: $[(a \vee b) \wedge (\neg a \vee c)] \rightarrow \neg(b \wedge c)$

The statement is false as written

Suppose a and c are true and b is false. Therefore it cannot be proved as a valid implication

$a=T,\ b=F,\ C=T$ is a counterexample

The valid resolution conclusion would be $b \vee C$

B

Clause	Expression	Reason
C_1	$p \vee q \vee \neg r \vee s$	Given
C_2	$\neg p \vee r \vee s$	Given
C_3	$\neg q \vee \neg r$	Given
C_4	$p \vee \neg s$	Given
C_5	$\neg p \vee \neg r$	Given
C_6	r	Given
C_7	$\neg q$	C_3, C_6; resolve on r
C_8	$\neg p$	C_5, C_6; resolve on r

C_9	$\neg s$	C_4, C_8; resolve on p
C_{10}	$p \vee q \vee s$	C_1, C_6; resolve on r
C_{11}	$p \vee s$	C_{10}, C_7; resolve on q
C_{12}	s	C_{11}, C_8; resolve on p
C_{13}	$\square$ (empty)	C_{12}, C_9; resolve on s

The empty clause is false under every assignment

The contradiction already follows from C_1, C_3, C_4, C_5, C_6; C_2 is not needed

C

Let $D_1 = p \vee \neg q$, $D_2 = \neg p \vee q$,

$D_3 = q \vee r$, $D_4 = q \vee \neg r$, $D_5 = \neg q \vee \neg r$

Choose $p = T$, $q = T$, $r = F$. Then

$D_1 = T \vee F = T$, $D_2 = F \vee T = T$,

$D_3 = T \vee F = T$, $D_4 = T \vee T = T$,

$D_5 = F \vee T = T$.

Since there are only $\boxed{5}$

disjunctions,

D.

Equations

$$x=y,\ y=z,\ z=-x \Rightarrow x=y=z=-x=0$$

Conversely, the assignment satisfies all three equations.

Propositional logic. Let

$$\phi = (A \to B) \wedge (B \to C) \wedge (C \to \neg A)$$

A	B	C	$A \to B$	$B \to C$	$C \to \neg A$	ϕ
T	T	T	T	T	F	F
T	T	F	T	F	T	F

A	B	C				
T	T	T	F	T	F	F
T	T	F	F	T	T	F
F	T	T	T	T	T	T
F	T	F	T	F	T	F
T	F	T	T	T	T	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

Exactly three assignments work:

$(A, B, C) = (F, T, T), (F, F, T), (F, F, F)$

Equivalently, $\phi \equiv \neg A \wedge (\neg B \vee C)$.